

Adopted Levels

$Q(\beta^-) = -1.595 \times 10^4$ 11; $S(n) = 17757$ 17; $S(p) = 83.6$ 5; $Q(\alpha) = -6563$ 3 [2021Wa16](#)
 $Q(\beta^-), S(n)$: Deduced by evaluators using mass excesses of 4777 105 for ^{35}Ca measured by [2023La09](#), and -1487 17 for ^{34}K measured by [2024Dr01](#); -11172.9 5 for ^{35}K from [2021Wa16](#). Values from [2021Wa16](#): $Q(\beta^-) = -16360$ 200 (syst), $S(n) = 18020$ 200 (syst). $S(2n) = 34860$ 200 (syst), $S(2p) = 4747.5$ 6, $Q(\varepsilon) = 11874.4$ 9, $Q(\varepsilon p) = 5978.2$ 5 ([2021Wa16](#)).
 Isotope discovery ([2012Th10](#)): $^{40}\text{Ca}(^3\text{He}, ^8\text{Li})^{35}\text{K}$ at Michigan State ([1976Be08](#)).
 ^{35}K decay measurements:
[1980Ew02, 1979Ca15](#): ^{35}K produced via $^{45}\text{Sc}(p, 8n3p)$ spallation at CERN. Measured $T_{1/2}$ and $\varepsilon + \beta^+$ -delayed protons and γ rays.
[2018Sa54, 2019ChZU](#): ^{35}K produced via $^1\text{H}(^{36}\text{Ar}, ^{35}\text{K})2n$ at Texas A&M. Measured $T_{1/2}$ and $\varepsilon + \beta^+$ -delayed protons and γ rays.
[1998Sc19](#): Polarized ^{35}K produced via fragmentation of ^{40}Ca on ^9Be target at GSI. Measured $T_{1/2}$ and g -factor of ^{35}K ground state from β -NMR.
[2006Me04](#): Polarized ^{35}K produced via $^{36}\text{Ar}(^9\text{Be}, ^{10}\text{Li})^{35}\text{K}$ at NSCL, MSU. Measured g -factor of ^{35}K ground state from β -NMR.
 ^{35}K mass measurements: [2023Zh10, 2007Ya08, 1976Be08](#).
 Theoretical calculations (binding energies, moments, radii, levels, J , π , mass, $T_{1/2}$, reaction rates, etc.): [2025Sh19, 2023Bo17, 2023Fo05, 2022Zo01, 2020Ma25, 2016Si02, 2016Me17, 2003Sm02](#).

 ^{35}K LevelsCross Reference (XREF) Flags

- A ^{35}Ca $\varepsilon + \beta^+$ decay (25.7 ms)
 B $^9\text{Be}(^{36}\text{Ca}, ^{35}\text{K})$
 C $^{40}\text{Ca}(^3\text{He}, ^8\text{Li})$

E(level) [†]	J^π [‡]	$T_{1/2}$	XREF	Comments
0.0	$3/2^+$	175 ms 2	ABC	$\% \varepsilon + \% \beta^+ = 100$; $\% \varepsilon p = 0.37$ 15 $\mu = (+)0.392$ 7 (2006Me04, 2019StZV) J^π : $L(^{36}\text{Ca}, ^{35}\text{K}) = 2$ from 0^+ and allowed $\varepsilon + \beta^+$ feeding to 4725.9, $1/2^+$ level in ^{35}Ar . Possible mirror level of $3/2^+$ ^{35}S g.s. $T_{1/2}$: weighted average of 175 ms 2 (2018Sa54), 178 ms 8 (1998Sc19), and 190 ms 30 (1980Ew02). μ : From β -NMR spectroscopy (2006Me04). Other: 0.36 3 (1998Sc19, \beta-NMR spectroscopy). The positive sign is based on the mirror ^{35}S g.s. $\% \varepsilon p$: from 1980Ew02 for $E(p) > 0.9$ MeV. $E(p) < 0.9$ MeV has also been observed (2018Sa54, 2019ChZU).
1553 5	$(1/2)^+$		A C	E(level): other: 1560 40 from $(^3\text{He}, ^8\text{Li})$. J^π : allowed $\varepsilon + \beta^+$ feeding from $3/2^+$ parent; possible mirror level of 1572, $1/2^+$ level in ^{35}S (1976Be08).
2690 50			C	E(level): from $(^3\text{He}, ^8\text{Li})$ only. Possible mirror level of 2717, $5/2^+$ level in ^{35}S (1976Be08).
3781 26	$1/2^+, 3/2^+$		A	
4018 37	$1/2^+, 3/2^+$		A	
4788 49	$1/2^+, 3/2^+$		A	
4982 13	$1/2^+, 3/2^+$		A	
5249 73	$1/2^+, 3/2^+$		A	
5533 49	$1/2^+, 3/2^+$		A	
5710 49	$1/2^+, 3/2^+$		A	
5865 38	$1/2^+, 3/2^+$		A	
6089 62	$1/2^+, 3/2^+$		A	
6335 73	$1/2^+, 3/2^+$		A	
9168 23	$1/2^+$		A	$T = 5/2$ J^π, T : isobaric analog state of $1/2^+$ ^{35}Ca g.s. (1985Ay01, 1999Tr04).

Continued on next page (footnotes at end of table)

Adopted Levels (continued)

 ^{35}K Levels (continued)

[†] From ^{35}Ca $\varepsilon+\beta^+$ decay, unless otherwise noted.

[‡] Allowed $\varepsilon+\beta^+$ feeding from $1/2^+$ ^{35}Ca parent, unless otherwise noted.